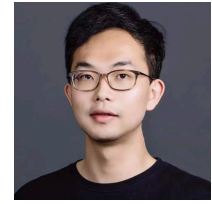


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Jiawang Bian is a Nanyang Assistant Professor at the College of Computing and Data Science, Nanyang Technological University (NTU), where he leads the Spatial Intelligence Lab. His research aims to enable AI to perceive, understand, and interact with the physical world—spanning 3D/4D reconstruction, generative AI, world models, and vision-language systems, with applications in autonomous driving, AR/VR, and robotics.

■ Education

Feb. 2019 – Apr. 2022 **Ph.D., Computer Science** at the University of Adelaide, Australia
Advised by Prof. Ian Reid and Prof. Chunhua Shen
Aug. 2012 – Jun. 2016 **B.Eng., Computer Science** at Nankai University, China
Advised by Prof. Ming-Ming Cheng

■ Positions

May 2026 – Present **Nanyang Assistant Professor**, NTU, Singapore
Apr. 2025 – May 2026 **Research Scientist**, ByteDance Seed, USA
Mar. 2024 – Apr. 2025 **Postdoctoral Researcher**, MBZUAI, UAE
Sep. 2022 – Dec. 2023 **Postdoctoral Researcher**, University of Oxford, UK
Aug. 2016 – Aug. 2017 **Research Assistant**, SUTD, Singapore

■ Research Internships

Sep. 2021 – Jan. 2022 **Research Intern**, Meta Reality Labs, UK
Apr. 2021 – Jul. 2021 **Research Intern**, Amazon Web Services, USA
Sep. 2018 – Jan. 2019 **Research Intern**, TuSimple, China

■ Grants & Fellowships

2026 – 2031 **NRF AI Fellowship** (Principal Investigator), National Research Foundation, Singapore
2026 – 2029 **Nanyang Assistant Professorship**, NTU, Singapore

■ Research Group

Postdocs Peng Yun (Ph.D., HKUST); Zihui Gao (Ph.D., Zhejiang University)
PhD Students Qihang Cao

Research Staff	Sensen Gao (Research Associate); Yuzhuo Tian (Research Assistant); Zehao Qi (Research Assistant)
Undergraduates	Yuchen Xie (NTU)

■ Teaching

Lecturer

2026 SC5006 Computer Vision & Applications, Co-Lecturer, NTU

Teaching Assistant & Guest Lecturer

2024 Advanced Research Methods, Teaching Assistant, MBZUAI

2022 Computer Vision, Guest Lecturer, University of Adelaide

2021 Master Data Science Research Project, Teaching Assistant, University of Adelaide

2020 Big Data Analysis and Project, Teaching Assistant, University of Adelaide

■ Open-source Impact

GMS GMS feature matcher: 1.1k+ GitHub stars; integrated into OpenCV

SC-Depth SC-SfMLearner and SC-Depth: 1.2k+ GitHub stars combined

SoL-Pi SoL-Pi (with NVIDIA): 3.3k+ GitHub stars

■ Professional Service

Journal Reviewer TPAMI, IJCV, TOG, TRO, TIP, TNNLS, TMM, RA-L, ISPRS, PR, KBS

Conference Reviewer NeurIPS, ICLR, SIGGRAPH, CVPR, ICCV, ECCV, AAAI, BMVC, ACCV, ICRA, IROS, WACV, PRCV, CVM

■ Publications

Over 40 publications in leading computer vision and machine learning venues, including CVPR, ICCV, ECCV, NeurIPS, ICML, ICLR, TPAMI, and IJCV; 7,000+ citations and an h-index of 23 (Google Scholar, Oct. 2026). Full list at jwbian.net. * Equal contribution; † corresponding author.

Selected Publications

- 1 W. Wang, Q. Cao, S. Gao, D. Y. Chen, H. Xu, W. Bian, S. Peng, T.-J. Cham, C. Zheng, A. Geiger, J. Cai, **J.-W. Bian**[†], B. Zhuang. Feed-Forward 3D Scene Modeling: A Problem-Driven Perspective. *arXiv* 2026. [\[arXiv\]](#)
- 2 S. Gao, Z. Wang, Q. Cao, D. Yu, C. Wang, T. Liu, M. Gong, **J.-W. Bian**[†]. OneWorld: Taming Scene Generation with 3D Unified Representation Autoencoder. *ECCV* 2026. [\[arXiv\]](#)
- 3 S. Gao, Z. Wang, Q. Cao, D. Yu, C. Wang, **J.-W. Bian**[†]. PixWorld: Unifying 3D Scene Generation and Reconstruction in Pixel Space. *arXiv* 2026. [\[arXiv\]](#) **280+ GitHub stars**

- 4 H. Liu, T. Ye, S. Gao, Q. Cao, Y. Li, M. Zhuge, D. Wang, R. Zhang, P. Luo, **J.-W. Bian**, L. Zhu, L. Zhu, E. Xie, S. Han. SoL-Pi: Recursively Scaling Auto-Research Loops for Efficient Agent Harness. *arXiv* 2026. [[arXiv](#)] **3.3k+ GitHub stars**
- 5 H. Li, Q. Cao, T. Tang, K. Xiang, Z. Guo, J. Han, **J.-W. Bian**, H. Xu, X. Liang. Thinking with Geometry: Active Geometry Integration for Spatial Reasoning. *ICML* 2026. [[arXiv](#)]
- 6 W. Wang, Y. Chen, Z. Zhang, H. Liu, H. Wang, Z. Feng, W. Qin, F. Chen, **J.-W. Bian**, Z. Zhu, D. Y. Chen, B. Zhuang. VolSplat: Rethinking Feed-Forward 3D Gaussian Splatting with Voxel-Aligned Prediction. *ECCV* 2026. [[arXiv](#)] **320+ GitHub stars**
- 7 Z. Gao^{*}, **J.-W. Bian**^{*}, G. Lin, H. Chen, C. Shen. SurfaceSplat: Connecting Surface Reconstruction and Gaussian Splatting. *ICCV* 2025. [[arXiv](#)]
- 8 J. Wu^{*}, **J.-W. Bian**^{*}, X. Li, G. Wang, I. Reid, P. Torr, V. Prisacariu. GaussCtrl: Multi-View Consistent Text-Driven 3D Gaussian Splatting Editing. *ECCV* 2024. [[Project](#)] **180+ citations**
- 9 **J.-W. Bian**, W. Bian, V. A. Prisacariu, P. H. Torr. PoRF: Pose Residual Field for Accurate Neural Surface Reconstruction. *ICLR* 2024. [[Project](#)]
- 10 G. Xu, W. Yin, J. Zhang, O. Wang, S. Niklaus, S. Chen, **J.-W. Bian**[†]. Towards Domain-agnostic Depth Completion. *Machine Intelligence Research* 2024. [[Paper](#)]
- 11 W. Bian, Z. Wang, K. Li, **J.-W. Bian**, V. A. Prisacariu. NoPe-NeRF: Optimising Neural Radiance Field with No Pose Prior. *CVPR* 2023. [[Project](#)] **CVPR Highlight** **440+ citations** **400+ GitHub stars**
- 12 L. Sun^{*}, **J.-W. Bian**^{*}, H. Zhan, W. Yin, I. Reid, C. Shen. SC-DepthV3: Robust Self-supervised Monocular Depth Estimation for Dynamic Scenes. *TPAMI* 2023. [[Paper](#)] **170+ citations**
- 13 **J.-W. Bian**, H. Zhan, N. Wang, T.-J. Chin, C. Shen, I. Reid. Auto-Rectify Network for Unsupervised Indoor Depth Estimation. *TPAMI* 2022. [[Paper](#)] **160+ citations**
- 14 **J.-W. Bian**, H. Zhan, N. Wang, Z. Li, L. Zhang, C. Shen, M.-M. Cheng, I. Reid. Unsupervised Scale-consistent Depth Learning from Video. *IJCV* 2021. [[Paper](#)] **SC-Depth series: 1,000+ citations** **1.2k+ GitHub stars**
- 15 **J.-W. Bian**, W.-Y. Lin, Y. Matsushita, S.-K. Yeung, T.-D. Nguyen, M.-M. Cheng. GMS: Grid-based Motion Statistics for Fast, Ultra-Robust Feature Correspondence. *CVPR* 2017. [[Paper](#)] **Integrated into OpenCV** **980+ citations** **1.1k+ GitHub stars**